Name :- Karthik.C
Reginton:- 192573008
Code :- CBA1501
Course :- Cloud Computing

CO - I

Assignment - I

1. Cloud computing :-

cloud computing is the delivery of computing services such as servers, storage, database, networking, software and analytics over the internet, allowing users to access resources on demand without owning physical infrastructure.

Characteristics :-

1. On - Demand self-Service :- Users can access computing resources whenever needed without human intervention.
2. Broad Network Access :- Services are available over the internet and can be accessed from laptops, smartphones, and tablets.
3. Resource pooling :- Resources are shared among multiple users using Virtualization.
4. Rapid Elasticity :- Resources can be increased or decreased quickly based on demand

Difference from On- Premise Computing :-

-> cloud does not require users to own (or) maintain hardware.

-> Resources are scalable and paid for based on usage.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution.

Several technology Advancements :-

Hardware Evolution :- Organizations initially relied on large mainframe computers. Later, personal computers and high-performance servers became common.

Networking Evolution :- The development of the internet enabled computers worldwide to communicate and share information efficiently.

Virtualization :- Virtualization technology allowed multiple virtual machines to run on a single physical server, improving resource utilization and reducing hardware costs.

Internet software Evolution :- Web applications enabled users to access software through browsers instead of installing it locally.

Modern cloud platforms :- These developments led to cloud platforms such as AWS, Microsoft Azure, and Google cloud, which provide scalable computing storage, networking and software services over the internet.

3. Explain the role of server Virtualization in cloud computing. Differentiate b/w Virtualization and cloud computing.

Server Virtualization is the process of creating multiple virtual servers on a single physical server using a hypervisor. Each Virtual machine operates independently with its own operating system and applications.

Role in cloud Computing

→ Maximizes hardware utilization
→ Reduces infrastructure cost
→ Allows quick deployment of virtual machines.
→ supports resource isolation and security.
→ Enables scalability and efficient resource management.

Differences:-

Virtualization

→ Technology that creates virtual machines
→ Focuses on hardware optimization
→ Runs inside physical servers
→ Foundation technology.

Cloud Computing

→ Service delivery model using the internet
→ Focuses on delivering IT services
→ Uses Virtualization plus networking and Automation.
→ Complete computing platform.

9. Discuss the role of data centers in delivering cloud services.

A datacenter is a secure facility containing servers, storage systems, networking equipment, cooling systems, and power backup that supports cloud computing.

Roles of Data Center

→ store customer data securely
→ Host cloud applications and websites
→ Provide continous service availability
→ Enable backup and disaster recovery
→ supports scalability by adding more servers when demand increases.
→ Ensure data reliable and high-performance cloud services.

Without data centers, Cloud computing services connot operate efficiently.

5 Differentiate Iaas, Paas, Saas, and Xaas with suitable real-world example for each model.

Service model

1. Iaas (Infrastructure as a Service)

-> Provides virtual servers storages, and networking resources.

Example :- Amazon EC2

2. Paas (Platform as a services)

-> Provides a platform for developing, testing and deploying applications.

Example :- Google App Engine

3. SaaS (software as a services)

-> provides ready-to-use software over the internet

Example :- Google Docs

4. Xaas (Anything as a Services)

-> Provides any IT services as an online service.

Example :- Microsoft 365.